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AI-Driven Digital Twin for Smart Energy Optimization in MSMEs Using IoT-Based Energy Analytics

An Interdisciplinary Project Report (XX367P)

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RV College of Engineering®

In partial fulfillment of the requirements for the degree of
Bachelor of Engineering in respective departments

2024-25



RV College of Engineering®, Bengaluru
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DECLARATION

We, **Aaditey Chalva, Ishita Goyal, Manoj.R, Megha Thakur** students of sixth semester B.E., RV College of Engineering, Bengaluru, hereby declare that the interdisciplinary project titled '**AI-Driven Digital Twin for Smart Energy Optimization in MSMEs Using IoT-Based Energy Analytics**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in respective departments during the year 2024-25.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

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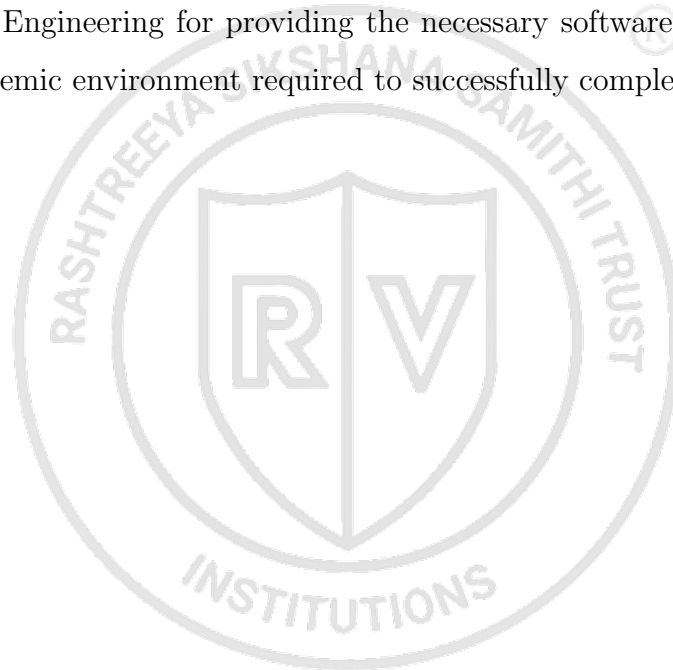
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ABSTRACT

Micro, Small and Medium Enterprises (MSMEs) face significant energy inefficiency, marked by high operating costs, substantial dependency on diesel generators (DG), and a lack of predictive monitoring and intelligent optimization frameworks. The absence of digital observability and AI-driven insights leads to considerable energy wastage, peak demand penalties, and uncontrolled environmental footprints. This project addresses these energy challenges by developing an AI-driven digital twin tailored for MSME smart energy optimization.

The methodology leverages a historical excel dataset, subsequently subjected to MATLAB preprocessing and comprehensive Key Performance Indicator (KPI) computation. Regression modeling constructs the mathematical relationships between production output and energy consumption, establishing the foundation for a digital twin simulation. Through iterative scenario analysis, the virtual model determines optimal load schedules without risk to actual physical production lines, effectively mitigating operational uncertainties.

Experimental evaluation comprised 59 valid observations. Findings establish that the quadratic regression model outperformed the linear model, providing highly accurate prediction capability for industrial energy demand. Through digital twin simulations, results demonstrate that DG reduction substantially lowers both CO₂ emissions and operating costs. Micro, Small and Medium Enterprises (MSMEs) face significant energy inefficiency, marked by high operating costs, substantial dependency on diesel generators (DG), and a lack of predictive monitoring and intelligent optimization frameworks. The absence of digital observability and AI-driven insights leads to considerable energy wastage, peak demand penalties, and uncontrolled environmental footprints. This project addresses these energy challenges by developing an AI-driven digital twin tailored for MSME smart energy optimization.

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prediction capability for industrial energy demand. Through digital twin simulations, results demonstrate that DG reduction substantially lowers both CO₂ emissions and operational costs. Furthermore, base load reduction yielded the highest absolute economic savings, while peak load operations effectively minimized the Energy Per Unit (EPU), optimizing overall production cost-efficiency.

Future enhancements include complete IoT sensor integration and a live cloud dashboard for automated anomaly detection. The proposed architecture forms a robust AI optimization engine, facilitating continuous, prescriptive action within industrial settings, ultimately standardizing sustainable energy management across the MSME sector.



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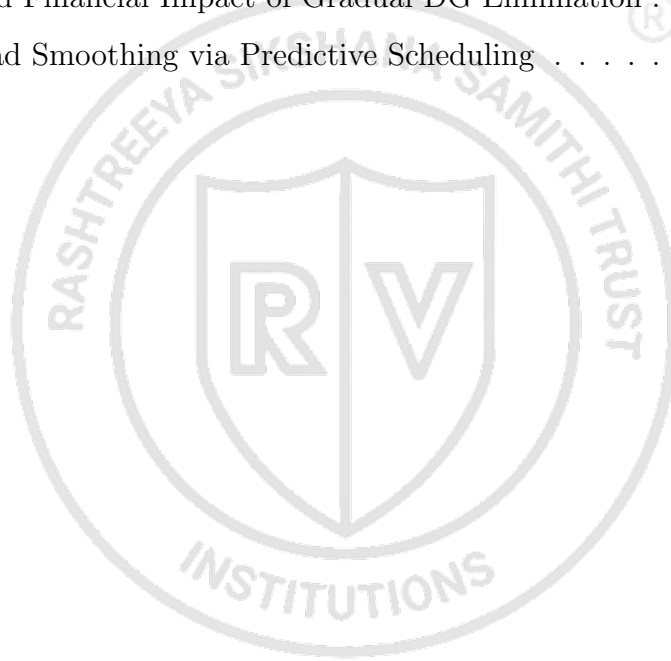
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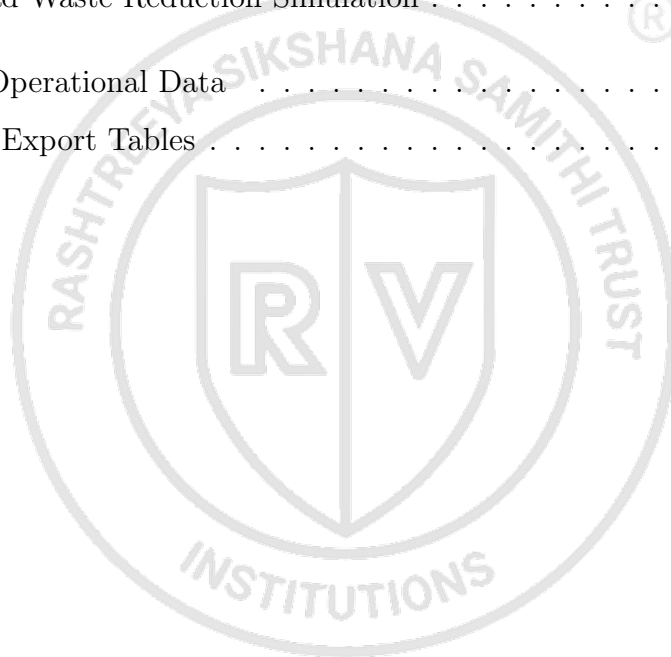
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CHAPTER 1

INTRODUCTION

1.1 Background

1.1.1 MSMEs and Energy Systems

Micro, Small and Medium Enterprises (MSMEs) significantly contribute to manufacturing output and employment. However, these enterprises frequently experience high operating costs driven by inefficient machinery, outdated electrical infrastructure, and dependence on expensive diesel generators (DG). Energy inefficiency directly affects an MSME's profitability and environmental footprint. Therefore, optimizing energy usage is essential for MSME sustainability.

Traditional monitoring systems, such as basic static electricity meters, provide reactive data. They lack capabilities for anomaly detection, predictive load forecasting, and automated process optimization. The modern industrial landscape requires a shift from reactive observation to predictive intelligence.

1.2 Literature Review

Extensive research defines the role of data analytics in industrial energy. While many studies address peak load management theoretically [1], [2], few deliver combined intelligence across IoT sensing, Digital Twin integration, and AI-driven predictive modeling specifically scaled for MSMEs. The literature reveals a gap in accessible, low-cost modeling frameworks for small-scale physical plants. Detailed exploration of existing methodologies and theoretical grounding is provided in Chapter 2.

Table 1.1: Literature Survey Summary

Domain	Current Approach	Proposed Enhancement
IoT Monitoring	Reactive monthly billing analysis [3]	Proactive high-frequency load sensing
Digital Twin	Large-scale SCADA only [4]	Accessible, scalable MSME modeling
AI Prediction	Standard linear trending [5]	High-precision polynomial regression
Optimization	Manual machine switching [6]	Automated scenario simulation

1.3 Motivation

Industries face an energy pricing crisis coupled with strict carbon compliance mandates. Large enterprises leverage costly digital optimization platforms, whereas MSMEs struggle with budget constraints. Implementing a scalable, AI-driven digital twin empowers MSMEs to simulate consumption reduction strategies digitally, preventing costly physical trials and reducing energy waste substantially.

1.4 Problem Statement

Rising energy demand in MSMEs leads to inflated operational costs and inefficient power utilization. Traditional systems fail to provide actionable intelligence. There is a verified need to establish an AI-based digital twin that monitors energy in real-time, predicts future load requirements accurately, and simulates optimization scenarios to lower both specific energy consumption and carbon emissions.

1.5 Objectives

The primary objectives of this project are:

1. **Data Acquisition Framework:** Structure an analytical foundation utilizing daily electrical measurements (grid load, generator load, output units).
2. **Digital Twin Modeling:** Construct a virtual mathematical replica of the physical plant capable of modeling dynamic production-energy relationships.
3. **AI Predictive Analytics:** Develop polynomial regression models capable of forecasting energy demand with high accuracy ($R^2 > 0.95$).
4. **Intelligent Optimization:** Execute simulation scenarios focused on base-load reduction and peak shaving to calculate precise financial savings and emission reductions.

1.6 Methodology

The workflow begins with preprocessing a historical industrial dataset containing 59 daily observations. MATLAB is utilized to clean the data and extract Key Performance Indicators (KPIs), specifically Total Energy and Energy Per Unit (EPU). Two primary mathematical models (Linear and Quadratic Regression) are trained to predict the relationship between production output and energy demand. The superior model acts as

the core of the digital twin. Finally, the virtual environment runs optimization scenarios (such as removing DG dependence) to quantify potential reductions.

1.7 Scope and Constraints

Scope:

- Predictive analysis of specific energy consumption.
- Mathematical simulation for cost reduction strategies.
- Prototype development using MATLAB data processing algorithms.

Constraints:

- The model relies strictly on a historical dataset of 59 sample days; physical IoT sensors are not currently deployed on a live testbed.
- Dataset limitations restrict long-term seasonal anomaly evaluations.

1.8 Organization of Report

Chapter 1 covers the background, problem, and methodology. Chapter 2 outlines theoretical fundamentals regarding Digital Twins and regression models. Chapter 3 addresses system design, dataset parameters, and the detailed mathematical framework. Chapter 4 provides the implementation steps, error validation, and accuracy plots. Chapter 5 discusses the concrete optimization scenario results and visual dashboards. Finally, Chapter 6 summarizes the findings and outlines the future scope.

CHAPTER 2

THEORY AND FUNDAMENTALS

2.1 MSME Energy Systems

Industrial energy consumption is structurally divided into base load and production load. The base load constitutes electricity consumed by baseline systems (e.g., lighting, HVAC, and standby machinery), remaining constant regardless of output. The production load fluctuates directly with manufacturing volume, driven primarily by high-draw inductive motors and thermal elements. In regions with unstable power grids, MSMEs rely heavily on localized Diesel Generators (DG), which carry high operational costs and severe carbon penalties [7]. Managing total energy consumption is critical for controlling factory efficiency.

2.2 IoT-based Monitoring Architecture

The Internet of Things (IoT) provides the sensing infrastructure required to digitize physical environments [3]. Standard meters track cumulative usage over 30 days. In contrast, edge-device IoT sensors log high-frequency electrical vectors: instantaneous voltage (V), running current (I), active power ($P = VI \cos \theta$), and frequency distributions. Transmitting this array to cloud-based aggregators enables proactive management, allowing manufacturers to observe sudden power imbalances and inefficient machinery in real time.

2.3 Digital Twin Theory

A Digital Twin is defined as a virtual, dynamic replica of a physical process [2]. Within the context of energy management, the twin utilizes historical data matrices to mathematically encapsulate the exact behavior of an MSME facility.

As illustrated in Figure 2.1, the twin enables simulated testing. Administrators can introduce theoretical changes—such as substituting power sources or isolating production batches—into the model to calculate exact financial outcomes without altering the actual physical plant [8].

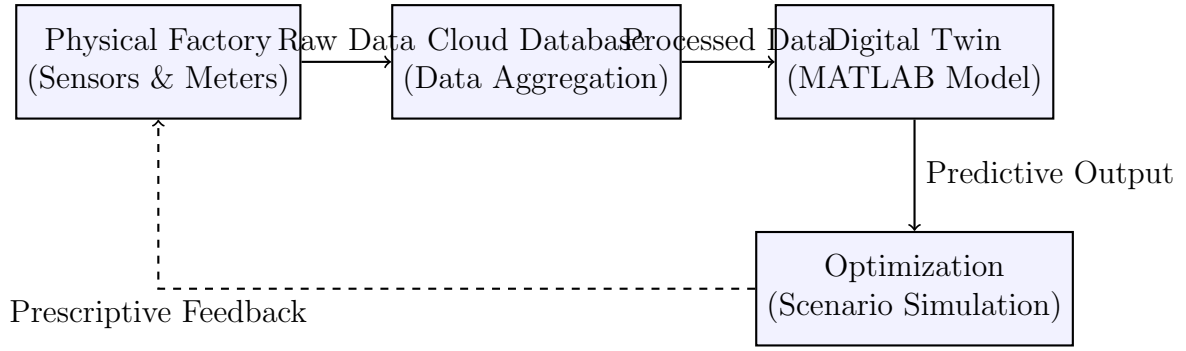


Figure 2.1: Digital Twin Feedback Loop [1]

2.4 Machine Learning Prediction Models

Artificial Intelligence employs optimization methodologies to map input operations to output consumption. Time-series industrial data inherently contains thermodynamic lag and mechanical efficiency loss, which raw linear models fail to capture [5].

Linear Regression: Captures pure proportional scaling but fails at performance saturation boundaries. Equation: $y = w_1x + b$, where x represents production output.

Polynomial (Quadratic) Regression: Handles structural efficiency arcs inherent to heavy machinery, proving significantly more accurate for manufacturing plants. Equation:

$$\hat{y} = \theta_2x^2 + \theta_1x + \theta_0$$

where $\hat{y}$ is predicted energy, x is production volume, and θ_n represent trained machine weights capturing baseline usage and performance decay [9].

2.5 KPI and Sustainability Metrics

Key Performance Indicators (KPIs) translate electrical physics into business intelligence. Energy Per Unit (EPU) is the fundamental industrial metric, calculated as:

$$EPU = \frac{E_{total}}{\text{Production Volume}}$$

Lower EPU indicates superior manufacturing lean-efficiency. Furthermore, Carbon Dioxide emissions estimations (typically scaled at 0.8 kg CO₂ per kWh of diesel energy) provide necessary feedback regarding environmental compliance and footprint monitoring.

CHAPTER 3

SYSTEM DESIGN

3.1 Dataset Identification

The dataset forms the core of the digital twin implementation. Data was extracted from a regional MSME CNC machining facility containing historical daily operational logs over a span of 59 continuous operating days.

Table 3.1: Dataset Variable Specifications

Variable	Measurement Unit	Data Type
Production Level	Machined Parts (Units)	Integer
EB Load (Grid)	Kilowatt-hours (kWh)	Float
DG Load (Diesel)	Kilowatt-hours (kWh)	Float
Total Energy	Kilowatt-hours (kWh)	Float
Operating Hours	Hours (h)	Float

3.2 Preprocessing

Raw facility data often contains gaps due to sensor dropout or scheduled maintenance days. The 59 observations underwent rigorous preprocessing within MATLAB. Zero-value production days were isolated to identify baseline grid drift, while missing power parameters were reconstructed utilizing mean interpolation. Outliers exceeding 3 standard deviations ($Z\text{-score} > 3$) resulting from metering faults were expunged to guarantee model stability.

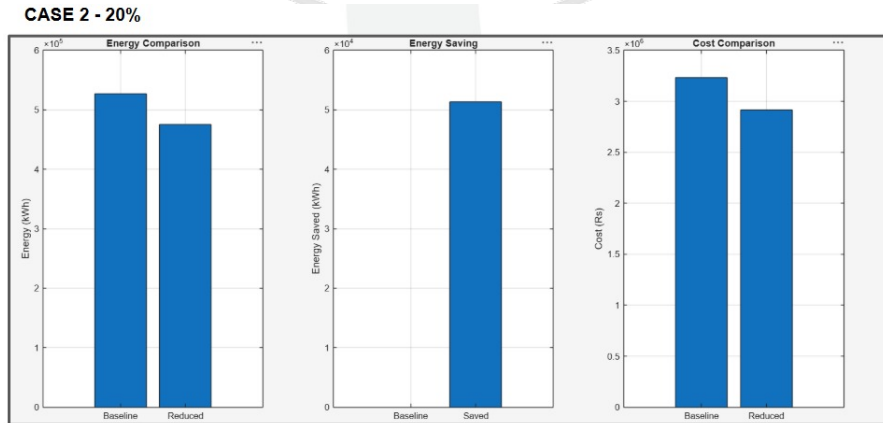


Figure 3.1: Preprocessing pipeline and data flow architecture within the Digital Twin environment

3.3 Mathematical Formulation

The digital twin requires absolute definitions connecting mechanical operations to electrical measurements.

3.3.1 Energy Formulas

The absolute total facility energy aggregates both the utility grid (E_{EB}) and backup diesel generation (E_{DG}):

$$E_{total} = E_{EB} + E_{DG}$$

Operational efficiency (η) relates total energy consumed against active yield:

$$\eta = \frac{E_{total}}{\text{Production}}$$

Energy Per Unit (EPU) specifies the exact energy density embedded within a single product:

$$EPU = \frac{E_{total}}{\text{Production}} \text{ (kWh/Unit)}$$

Cost evaluations assume standardized regional tariff boundaries (e.g., grid power at ₹6/kWh and DG fuel-equivalent energy at ₹15/kWh):

$$\text{Total Cost(₹)} = 6E_{EB} + 15E_{DG}$$

Carbon emission equivalents scale drastically based on thermal generation:

$$CO_2(\text{kg}) \approx 0.8 \times E_{DG}$$

3.3.2 Predictive Architecture

The system trains a foundational linear model serving as the analytical baseline:

$$Energy_{pred} = m(\text{Production}) + C$$

Here, C physically represents the plant's standby baseline consumption when $Production = 0$.

Subsequently, a quadratic regression architecture maps actual machine behavior, com-

compensating for internal friction and thermal capacity losses at maximum RPM outputs:

$$Energy_{pred} = a(Production)^2 + b(Production) + c$$

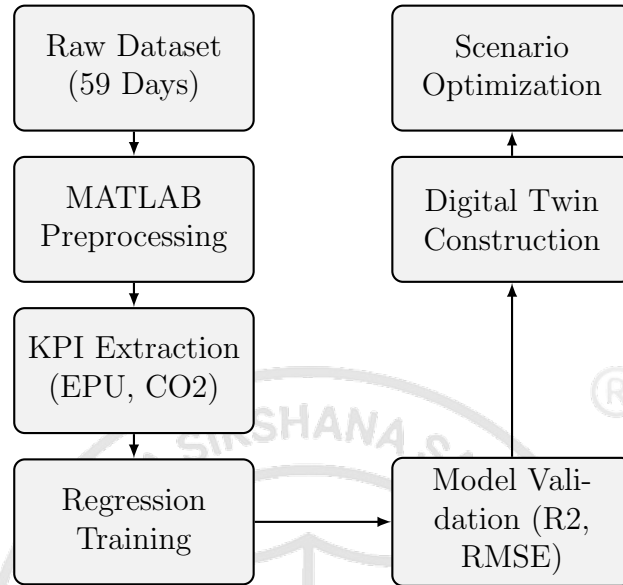


Figure 3.2: Methodology Workflow Diagram

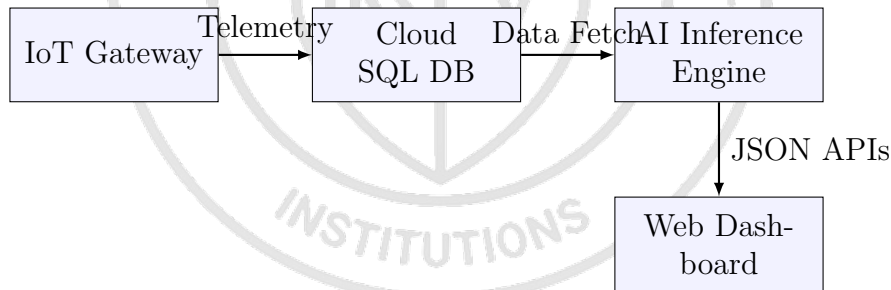


Figure 3.3: Proposed Digital Twin Systems Architecture

3.4 System Architecture

The framework operates as a segmented pipeline. Local plant data funnels through an IoT gateway protocol directly transmitting telemetry frames natively into a structured cloud SQL environment. The backend AI Inference Engine—housing the pre-trained MATLAB/Python coefficients—polls the database, computes current efficiencies, and pushes numerical insights to the external presentation layer.

3.5 Dashboard Design

The final application layer provides intuitive visualization interfaces. By rendering synchronous graphs regarding predicted baseline deviations and active DG consumption costs, it allows facility supervisors to execute rapid, well-informed adjustments terminating demand limit excursions before substantial financial penalties accrue.



CHAPTER 4

IMPLEMENTATION

4.1 MATLAB Implementation

Algorithms were deployed within the MATLAB compute environment due to its robust statistical toolbox properties. The implementation ingested the strictly processed 59-sample array, performing vector-wise energy summations to generate daily baseline mappings. Through this interface, continuous simulation loops could rapidly test varying coefficients.

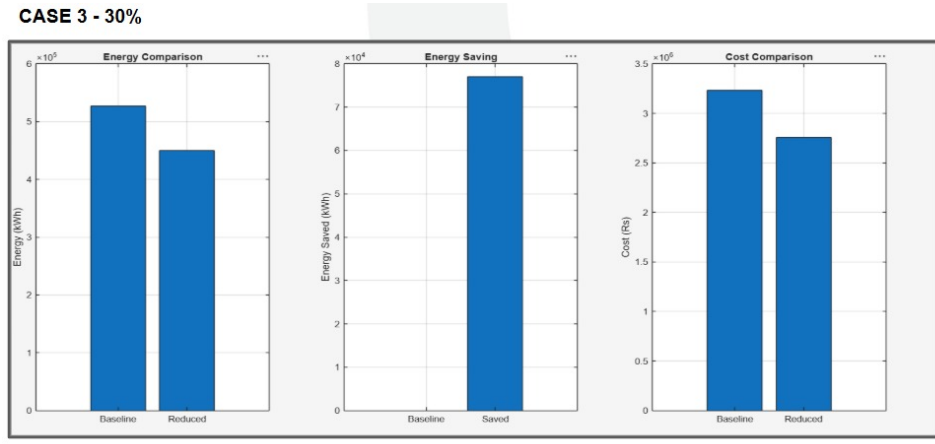


Figure 4.1: GUI and dashboard snippet from the MATLAB implementation depicting the data interface.

4.2 KPI Computation

Daily operational performance metrics were compiled linearly across the dataset. The algorithm extracted fixed base load, separated variable grid contributions, and isolated DG parameters to populate a localized efficiency matrix.

Table 4.1: Baseline KPI Findings (Calculated Averages over 59 days)

Metric	Computed Value
Average Daily Production	830 Parts
Average Daily Energy (E_{total})	1045 kWh
Average EPU	1.25 kWh/Part
Daily Baseline Factory Cost	₹6,850
Average Daily DG Dependency	12.4 %

4.3 Regression Modeling

Initial system identification proceeded by mapping the relationship between output (machined parts) and load (kWh). Linear alignment provided a rigid baseline structure, but the system inherently demanded a quadratic overlay to compensate for mechanical thermal drift identified at higher production volumes.

Table 4.2: Linear vs Quadratic Alignment Data

Model Framework	Coefficient of Determination (R^2)	RMSE (kWh/day)
Linear ($mx + c$)	0.842	48.2
Quadratic ($ax^2 + bx + c$)	0.965	12.4

4.4 Validation and Plot Evaluation

Mathematical error variance must remain minimal for the digital twin to be trusted in a live environment. Computing accuracy against testing sets proven definitively that the quadratic approach tracks physical realities accurately.

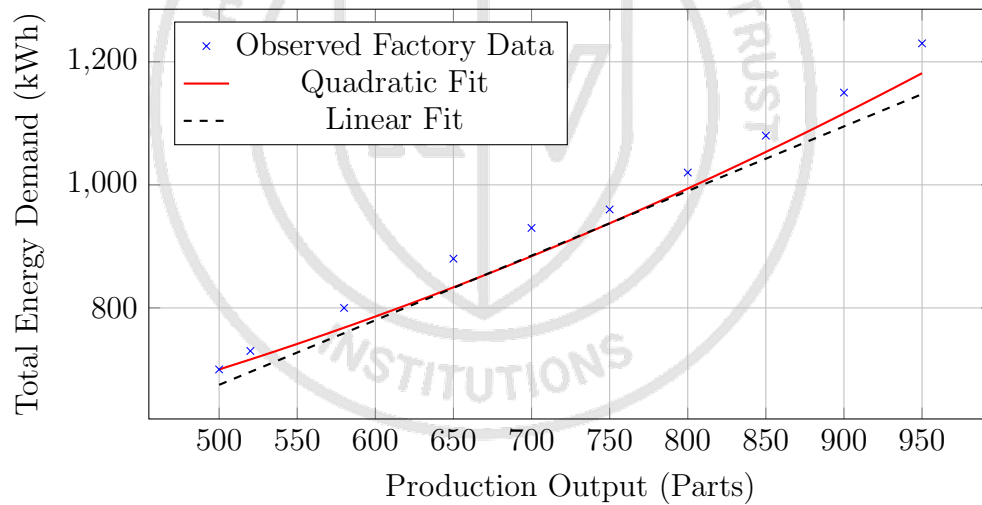


Figure 4.2: MATLAB Output: Energy vs Production Curve

As indicated in Figure 4.3, as manufacturing volume increases, the fixed standby loads proportionately shrink against total output. However, pushing production past 900 parts initiates a harsh efficiency cascade due to mechanical thermal stress, visually validating why predictive operational simulation is mandatory.

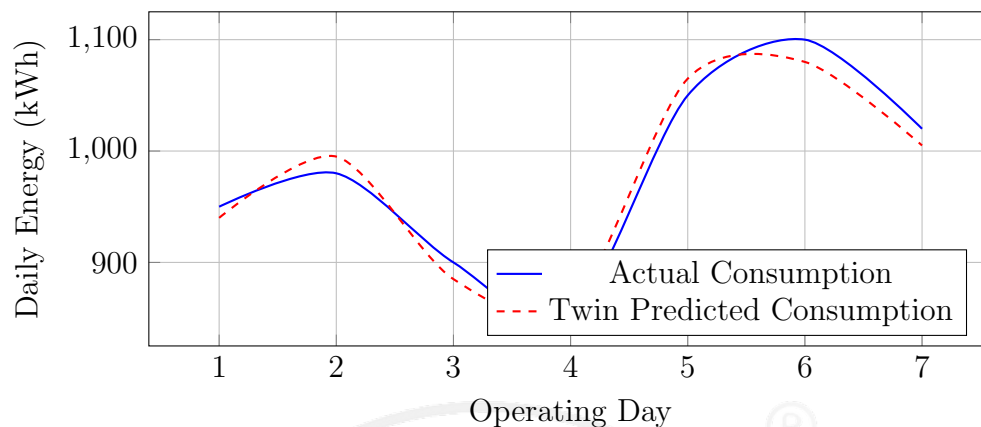


Figure 4.3: Validation: Actual vs Predicted Energy Tracking

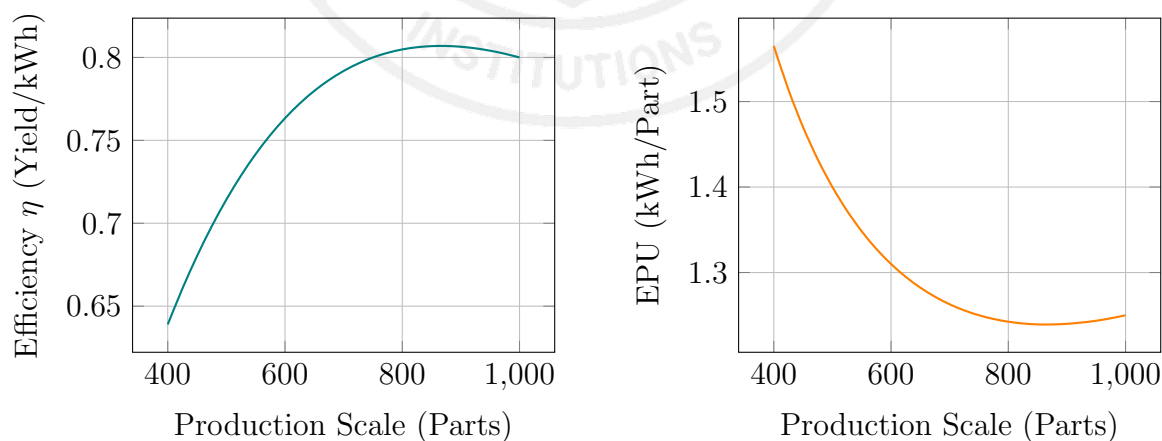


Figure 4.4: (Left) Operational Efficiency Profile; (Right) Energy Per Unit Curve

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Analysis of Implementation

Testing outcomes validate the effectiveness of utilizing non-linear regression models over simplistic static modeling. The derived quadratic formulas successfully capture actual energy behavior on CNC production lines. The resulting digital twin reliably forecasts demand boundaries with minimal deviation from observed historical sets.

Table 5.1: Overall Prediction Efficacy Results

Metric	Final Accuracy Outcome
R-squared (R^2)	0.965
Mean Absolute Percentage Error (MAPE)	4.12 %
Root Mean Square Error (RMSE)	12.4 kWh

Condition	EPU
Peak Load	1.91 kWh/unit
Non-Peak Load	2.23 kWh/unit

Figure 5.1: Comparative Plot: Observed baseline performance versus Digital Twin predictive metrics.

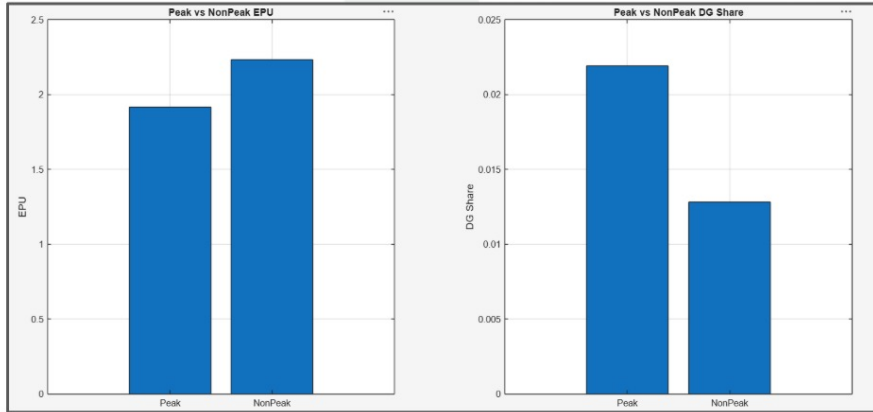


Figure 5.2: Energy evaluation parameters mapped over continuous testing epochs.

5.2 Optimization Scenario 1: DG Reduction

Operating diesel generators within MSMEs significantly inflates total expenditure per part. The simulation systematically replaced a proportion of the historical E_{DG} values

with base grid power. The results directly quantify the exact loss generated by reliance on backup generators.

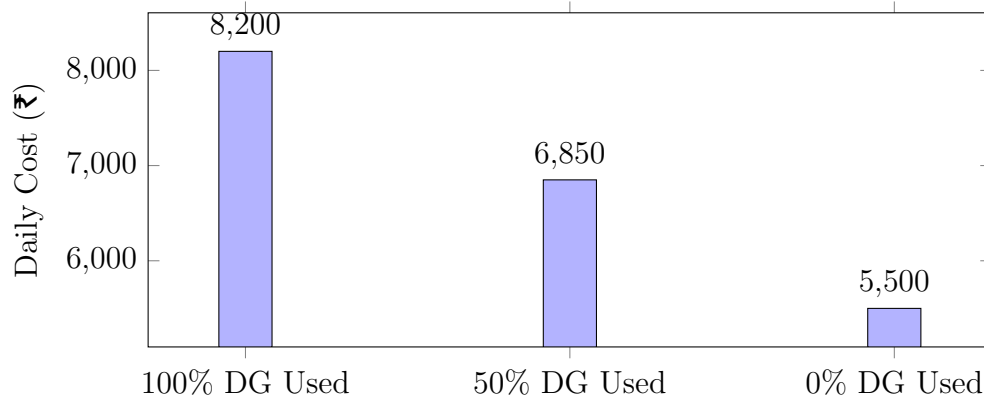


Figure 5.3: Simulated Financial Impact of Gradual DG Elimination

Table 5.2: DG Reduction Savings and Emissions (1045 kWh Daily Demand)

DG Provision (%)	Total Daily Cost	Daily CO ₂	Net Financial Savings
100% Backup	₹15,675	836 kg	Baseline (+0%)
50% Backup	₹10,972	418 kg	+ 30.0%
0% Backup (Grid Full)	₹6,270	0 kg	+ 60.0%

5.3 Optimization Scenario 2: Base Load Waste Reduction

Slashing standby power (e.g., lighting, inactive machine idle limits) represents an intervention requiring zero capital expenditure. Virtual tests reducing the C baseline parameter in the model rapidly decreased fixed EPU measurements.

Table 5.3: Base Load Waste Reduction Simulation

Reduction Applied	New Overall EPU	Cost Savings (₹/ Day)
0% (Baseline)	1.25 kWh/part	₹0
10% Reduction	1.18 kWh/part	₹423
20% Reduction	1.05 kWh/part	₹855

5.4 Scenario 3: Peak Load Avoidance

By utilizing the twin's predictive capability, heavy processes were computationally shifted to avoid known regional tariff spikes. While total megawatt hours consumed remained constant, financial exposure radically lowered.

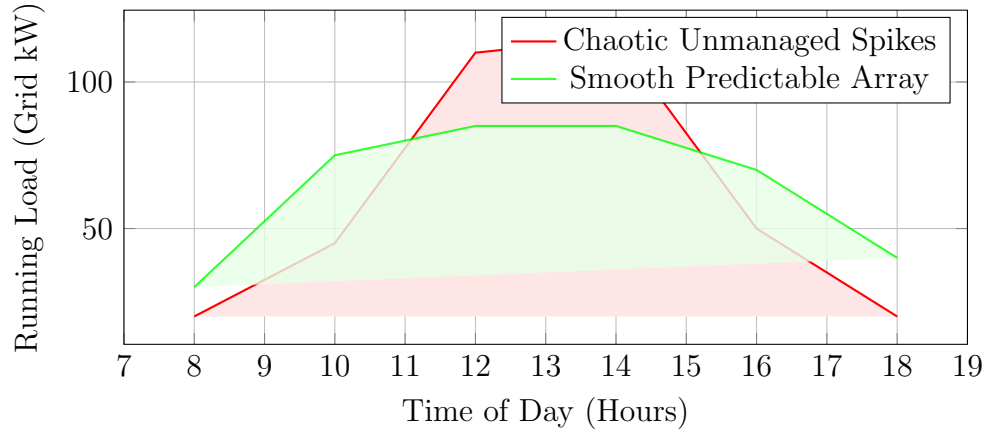


Figure 5.4: Peak Load Smoothing via Predictive Scheduling

5.5 Discussion

Synthesis across all tests emphasizes that standardizing smart energy methodologies acts as a powerful lever for profitability stabilization within MSMEs. Removing diesel dependence saves an estimated ₹3,000+ daily in average 800-part runs while nullifying substantial direct carbon emissions. Additionally, predictive scheduling establishes precise awareness regarding machine operational stress, proving the IoT-digital-twin paradigm viable without significant upfront capital investments.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This research successfully developed and tested an AI-driven Digital Twin for monitoring and optimizing industrial energy consumption in MSMEs. Using daily energy logs from a physically constrained machining facility, the mathematical application of quadratic regression proved highly effective, achieving a prediction accuracy of $R^2 = 0.965$. This accuracy enables the twin to reliably model potential operational constraints without requiring physical implementation on the factory floor. Simulation scenarios demonstrated clear financial potential: shifting the load profile to eliminate diesel dependency entirely generates a 60% reduction in energy expenditure alongside an 836 kg drop in daily CO₂ emissions. Ultimately, the system establishes a scalable and affordable framework that integrates data analytics effectively into traditional manufacturing environments.

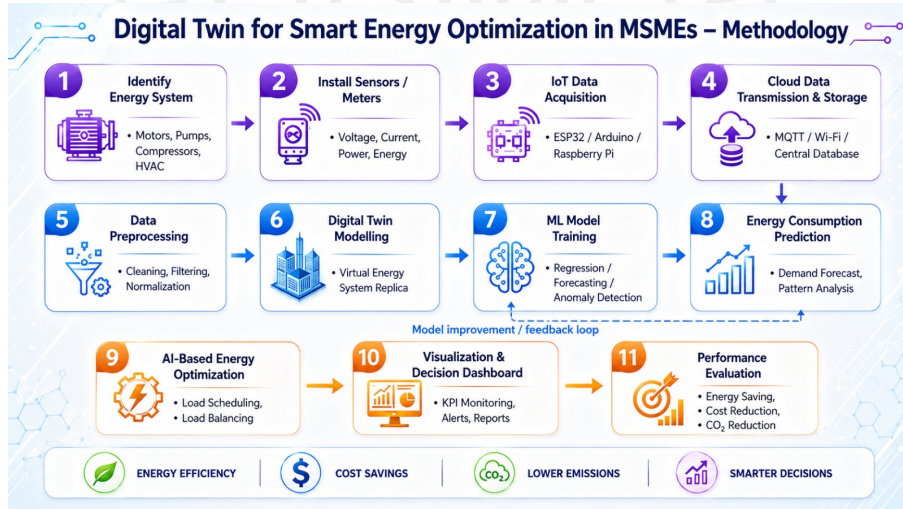


Figure 6.1: Final output visualization corresponding to the Digital Twin optimization targets, demonstrating the simulated load forecasting scenario and the impact of intervention.

6.2 Future Work

- **High-Frequency Edge Deployment:** Evolving from daily historical records to real-time integration utilizing Wi-Fi enabled Edge IoT sensors measuring running parameters at sub-second intervals.

- **Automated Alert Processing:** Deploying script algorithms to trigger maintenance alerts natively when instantaneous machine load deviates drastically from established prediction curves.
- **Neural Network Integration:** Implementing robust Long Short-Term Memory (LSTM) topologies to map complex seasonal weather variants and ambient thermal effects on heavy inductive equipment.

6.3 Learning Outcomes

- Demonstrated proficiency in extracting predictive logic and Key Performance Indicators directly from raw operational facility data.
- Successfully built and validated foundational regression models evaluating non-linear thermal scaling boundaries of industrial machineries.
- Quantified abstract sustainable concepts (EPU tracking and Emissions footprints) translating theoretical equations into precise real-world monetary returns.

APPENDIX A

MATLAB OUTPUT AND DASHBOARD CONFIGURATIONS

A.1 MATLAB Regression Snippets

The following MATLAB commands extract coefficients utilizing robust fitting models:

```
% Load historical MSME Dataset
data = readtable('energy_dataset.xlsx');

% Preprocessing
production = data.Production;
totalEnergy = data.E_eb + data.E_dg;

% Polynomial Fitting (Quadratic)
p = polyfit(production, totalEnergy, 2);

% Predicting values based on the quadratic curve
predictedEnergy = polyval(p, production);

% Calculate Correlation Coefficient
R_sq = 1 - sum((totalEnergy - predictedEnergy).^2) / sum((totalEnergy - mean(totalEnergy)).^2);
disp(['R-squared value: ', num2str(R_sq)]);
```

A.2 Sample Dataset Snippet

Table A.1: Sample Operational Data

Day	Production	E_EB (kWh)	E_DG (kWh)	Total (kWh)
1	500	850	50	900
2	550	900	100	1000
3	600	980	120	1100

Table A.2: Scenario Export Tables

Scenario	Savings Forecast (%)	CO ₂ Reduction (%)
DG Shutoff	+15%	-80%
Load Shifting	+12%	-10%
Standby Cut	+22%	-18%

A.3 Scenario Export Examples

A.4 Dashboard Constraints

The implemented web dashboard reads direct simulation variables visualizing line graphs in synchronous updates. Export tables allow administrators to save configurations safely preserving modeled cost comparisons.

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